

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

**COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems**
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I C Dilip Kumar (192412133) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C Dilip kumar

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

5. Edge AI for Intelligent 5G Network Optimization

The report presents a study of Edge AI for intelligent optimization of 5G networks, focusing on the use of Artificial Intelligence (AI) and Machine Learning (ML) at Multi-Access Edge Computing (MEC) nodes to improve network performance. Edge AI brings intelligent processing closer to users and 5G base stations, enabling faster decision-making, reduced latency, efficient resource utilization, and improved Quality of Service (QoS).

The report explains the integration of 5G, MEC, and Edge AI architecture, consisting of 5G User Equipment (UE), gNodeBs, edge computing nodes, AI/ML models, and centralized cloud infrastructure. Edge AI models can process real-time network information such as traffic load, SNR, channel conditions, latency, throughput, packet loss, and user demand. The AI system can then make intelligent decisions for resource allocation, traffic management, network slicing, and service placement.

Network optimization is an important challenge in 5G because the network must support diverse services such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and massive Machine-Type Communication (mMTC). Conventional static resource-allocation techniques may not respond effectively to rapidly changing traffic and channel conditions. Edge AI enables dynamic and adaptive optimization by continuously learning from network conditions and predicting future resource requirements.

The report discusses AI techniques such as Deep Learning, Reinforcement Learning (RL), and Deep Reinforcement Learning (DRL) for intelligent network management. These techniques can be used for radio-resource allocation, congestion control, user association, load balancing, network slicing, mobility management, and edge-server selection. Reinforcement learning is particularly useful because an intelligent agent can learn suitable resource-allocation policies based on network states and QoS requirements.

Performance is mainly evaluated using parameters such as latency, throughput, packet loss, energy efficiency, resource utilization, and QoS satisfaction. Edge AI can reduce decision-making latency because data does not always need to be transferred to a remote cloud server. Intelligent resource allocation and traffic prediction can also improve throughput and prevent network congestion. However, Edge AI introduces challenges related to computational complexity, training-data requirements, model accuracy, privacy, security, and energy consumption at edge nodes.

A key finding of the report is that Edge AI and 5G can complement each other to create an adaptive and intelligent communication network. Real-time network monitoring combined with AI-based decision-making allows resources to be dynamically adjusted according to traffic demand and channel conditions. The integration of MEC, AI/ML, network slicing, SDN, and NFV provides a promising framework for autonomous 5G network optimization.

In conclusion, Edge AI is an important technology for achieving intelligent, low-latency, and efficient 5G network operation. Deploying AI models closer to the network edge enables faster optimization decisions, improved resource utilization, higher throughput, and better QoS. Future research should focus on federated learning, explainable AI, intelligent network orchestration, distributed reinforcement learning, energy-efficient AI models, and secure Edge AI for beyond-5G and 6G networks.

Reference:

G. Nencioni, R. G. Garroppo, and R. F. Olimid, "5G Multi-Access Edge Computing: A Survey on Security, Dependability, and Performance," *IEEE Access*, vol. 11, pp. 63496–63527, 2023.



Edge AI for Intelligent 5G Network Optimization: A Survey on Techniques, Applications, and Challenges

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ABSTRACT The Fifth Generation (5G) mobile network is expected to support diverse services with stringent performance requirements. Efficient network optimization is essential to meet the demands of high data rates, ultra-low latency, massive connectivity, and dynamic traffic patterns. Edge Artificial Intelligence (Edge AI) empowers network nodes at the edge to perform real-time analysis and intelligent decision-making, enabling proactive and adaptive network optimization. This paper presents a comprehensive survey of Edge AI techniques for intelligent 5G network optimization. We review machine-learning and deep-learning methods, including reinforcement learning, deep reinforcement learning, federated learning, and graph-based learning, and their applications in resource allocation, traffic prediction, load balancing, network slicing, mobility management, and congestion control. We discuss relevant datasets, performance metrics, and benchmarking results reported in the literature. Finally, we highlight open challenges and future research directions for secure, dependable, and energy-efficient Edge AI-enabled 5G and beyond-5G/6G networks.

INDEX TERMS 5G, Edge AI, Multi-Access Edge Computing, Machine Learning, Reinforcement Learning, Network Optimization, Resource Allocation, Network Slicing, Intelligent Networks.